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Pediatric Dental Anesthesia: A Comprehensive Analysis of Heart Rate Response, Oxygen Saturation, and Patient Comfort with the STA System

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0. Abstract

This study aimed to develop a safer and more effective protocol for local anesthesia in pediatric dentistry using the Single Tooth Anesthesia (STA) system, addressing concerns about cardiac events during procedures. By analyzing the interactions between heart rate, oxygen saturation, and patient comfort, we sought to enhance pediatric dental care.

Data was collected from pediatric dental procedures in a private clinic, with vital signs recorded via pulse oximetry and anesthesia details captured through the STA system. The analysis revealed subtle differences in maximum heart rate (HR) across different anesthetic types and methods, a slight negative correlation between patient age and HR, and significant HR fluctuations associated with discomfort events like crying and gagging. Additionally, variability in anesthesia dosage requirements across methods highlighted the need for precise dosing.

The findings emphasize the importance of selecting anesthesia methods that minimize cardiac risks, maintain stable physiological conditions, and enhance patient safety. Further data collection and analysis are recommended to refine these protocols and improve outcomes in pediatric dentistry.

1. Introduction

Dental anesthesia is an important tool in modern dentistry, providing patients with comfort and pain relief during dental procedures which often include the removal of gum tissue, cavity preparation and tooth extractions. In addition to immediate pain relief, a less painful procedure results in increased patient compliance with dental visits, improving the patient's dental hygiene which is extremely important in pediatric dentistry [1,2]. Although typical dental anesthetics are generally safe and have been shown to maintain cardiovascular functions in patients [1,3,4], the few rare occurrences of pediatric patients having cardiac events linked with the anesthetics used in the procedure that are rising over the years, question their safety [5,6,7].

1.1 Anesthetics

An anesthetic is a substance used in dental practice to anesthetize the area where dental procedures are performed, their purpose is to block nerve impulses, which reduce or eliminate the sensation of pain. This is essential for making dental treatments more comfortable and allowing dentists to work effectively without causing undue discomfort to patients [8]. Even with anesthetics, some discomfort can still be present during dental procedures. Patients may express discomfort through reactions such as flinching, clenching their fists, tensing their muscles, grimacing, or vocalizing feelings of

pressure, discomfort, or pain. Some patients may also experience involuntary reflexes like gagging or tearing up due to the sensation of the instruments or pressure applied during the procedure.

1.2 CHEOPS

CHEOPS (Children's Hospital of Eastern Ontario Pain Scale) is a tool used to quantify pain and discomfort in pediatric patients, that has been shown useful as pain quantifier in children [9,10]. It assesses factors like facial expression, verbalization, leg movement, activity level, and crying to determine pain levels. Healthcare providers use CHEOPS to better understand and address the needs of children experiencing pain, especially when verbal communication may be limited [11-13]. This understanding allows dentists to evaluate scientifically the discomfort of certain procedures and anesthetic methods.

1.3 Local Anesthetic methods

Three commonly used injection techniques to administer local anesthetics are Infiltration anesthesia, local nerve blocks, and sulcular anesthesia.

Infiltration is usually used in the maxilla and involves the injection of local anesthetic into the tissues surrounding the teeth of the procedure, the anesthetic diffuses through the tissue, resulting in localized numbness of the area [8,14,15].

Local nerve block involves injecting local anesthetic around specific nerves to block a specific nerve, blocking sensation in a larger area [16]. There are three main types of nerve blocks: buccal, lingual, and main nerve block. Buccal block numbs the buccal nerve in the injected side of the mouth. The nerve supplies sensation to the posterior buccal mucosa and the gums of the second and third mandibular molars [8,17,18]. Lingual block anesthetize the lingual nerve, which supplies sensation to the anterior two-thirds of the tongue, the floor of the mouth, and the tongue side gums [8,17,18].

Main nerve block usually refers to a combination of inferior alveolar nerve blockage, which supplies sensation to the pulp and periodontium of all the molar and premolar teeth on one side of the mouth, as well as infiltration to the buccal and lingual nerves [17,18]. Sulcular anesthesia involves applying local anesthetic directly into the gingival sulcus, the space between the treated tooth and the surrounding gum tissue, resulting in more localized anesthesia [8,19].

The choice of anesthetic method mostly depends on the number of procedures in the session, the number of teeth involved, location and what teeth are involved.

For dosage calculation, dentists typically use guidelines based on the patient's weight, age, medical history, and the type of procedure being performed. Various online calculators and software tools are available to help determine the appropriate dosage of local anesthetic based on these factors [8], ensuring safe and effective pain management during dental procedures.

1.4 Local Anesthetic substances

Three widely used local anesthetics in dentistry with the mentioned anesthesia methods are Lidocaine, Mepivacaine and Articaine. by blocking the sodium channels in the nerve, preventing signals from passing through them, causing numbness [20-22]. Lidocaine has a fast onset of action, between 1-5 minutes, but works for short periods of time, Articaine works immediately after injection due to its enhanced ability to penetrate soft tissues, also for a brief period, and Mepivacaine has slower onset of action, usually around 15 min, but has longer duration [20-22]. Although studies show that all three local anesthetics are safe and effective in dental use, the studies favor the use of Articaine over Lidocaine for short procedures and support the use of Mepivacaine in patients with prior cardiovascular conditions [20-27]. Each of these local anesthetics has its own advantages and considerations regarding onset, duration, injection pain, allergies and effectiveness, and dentists may choose the most appropriate option based on the specific needs of the patient and the dental procedure being performed [8].

1.5 Adrenaline and with local anesthetics

To enhance the effects of the anesthetic used, Adrenaline is commonly added to local anesthetics used in dentistry. Adrenaline is a natural hormone released from the adrenal gland and mostly causes vasoconstriction, which reduces blood flow to the area where the anesthetic is injected [28,29]. This helps to prolong the anesthetic effect by slowing down the absorption of the anesthetic into the bloodstream [29,30]. As a result, the anesthesia lasts longer, allowing the dentist to perform the procedure more effectively without needing to administer additional doses of anesthetic [30]. In addition to extending the duration of the anesthetic, the vasoconstriction property also helps to reduce bleeding at the injection site and in the surrounding tissues during dental procedure, making it safer and more visible [30]. The injected adrenaline is more prominent in the injection site which often diffuses into the bloodstream throughout the body, activating the sympathetic system, causing blood pressure to elevate and heart rate to rise [30]. It is important to monitor the heart rate of patients under anesthesia with and without adrenaline included anesthetics to detect any allergic reaction, abnormal heart rate and oxygen saturation deficiency which can be cheaply and non-invasively detected with a pulse oximeter [30-33].

1.6 Pulse oximetry in dentistry

A pulse oximeter is connected to the finger of the patient and by sending 2 wavelengths of light, blue and red, and by reading the deflection of each one of them, can accurately estimate the blood oxygen saturation due to slight difference in color between oxygenated and deoxygenated blood. [34,35]. In addition, it has a pressure sensor that can detect heart pulses, calculate the intervals between them, and measure the heart rate [34]. Heart rate is also an indicator for the function of the autonomic nervous system which reflects negative emotions and shows if one is anxious or relaxed. It is recommended to monitor the heart rate while injecting anesthetic, especially in pediatric patients which are more susceptible to dental anxiety [31,36,37].

1.7 STA system

The purpose of dental anesthesia administration is to decrease the amount of pain during procedures. The injection itself however, might be the most prominent cause of pain as well as the main focus of dental anxiety among patients [37]. To avoid injection pain and the fear of anesthetic injection, the Single Tooth Anesthesia (STA) system was invented. STA is a technique used in dentistry to administer local anesthesia precisely to a single tooth or a small area of the mouth, targeting only the specific nerve endings responsible for sensation in that area. This approach allows for more targeted and effective pain control, reducing the need for widespread anesthesia and minimizing post-procedural numbness in adjacent tissues [38-40].

Figure 1: Single-use needle for the STA [40]

The STA system typically consists of a computer-controlled injection device that delivers a precise and controlled amount of local anesthetic solution through a fine needle, as can be seen in figure 1. The device utilizes pressure-sensing technology to monitor tissue resistance in real-time, ensuring accurate delivery of anesthesia without causing discomfort to the patient. By delivering anesthesia directly to the nerve endings associated with the targeted tooth, STA minimizes the risk of systemic side effects and provides more predictable and efficient pain management during dental procedures [38-40].

2. Objectives

Lack of information on dental deaths and the focus of existing research in pediatric dental anesthesia on conventional protocols that do not involve the STA method leave a gap in understanding the dynamic interplay between heart rate response, oxygen saturation, and patient comfort with or without the STA system [41-43]. This research focuses on developing a novel protocol for local anesthesia administration using the STA system in pediatric dentistry by evaluate the dynamics of heart rate response, oxygen saturation, and patient comfort during the administration of local anesthesia. By analyzing these factors, our study seeks to enhance patient care and safety in pediatric dental practice through the introduction of a new anesthesia protocol and shed light on the rise in dental cardiac events.

3. Materials and Methods

3.1 Research data acquirement

Data was collected from the private pediatric dentistry of Ashkenazi Malka in Petah Tikva, Israel, from procedures where local anesthesia was administered. Data was collected using a pulse oximeter (Nellocar N550, Boston, USA) with sampling frequency of 0.5 [Hz], STA system (computerized controlled local anesthesia delivery – Single-Tooth-Anesthesia, Milestone Scientific, Inc. Deerfield, IL, USA) with a sampling frequency of 6 [Hz], and manually using data recording software that was developed by previous student projects. The recorded data was then sent to us as a Microsoft Access database.

Date of birth, height, weight, BMI, teeth sedated, teeth treated, anesthetic method, anesthetic substance, adrenaline ratio and discomfort events were recorded manually to the software, the amount of anesthetic used and total number of ampules was recorded by the STA system, and the heart rate and oxygen saturation were recorded using the pulse oximeter.

3.2 Data Segmentation & Processing

3.2.1 Data Segmentation

The collected data was segmented based on the identified correlations and relationships of interest. Each correlation, such as the relationship between patient age and maximum blood pressure or the association between anesthesia dosage and signs of patient discomfort, was categorized to facilitate focused analysis. This segmentation process ensured that the data was organized effectively for subsequent processing and analysis. Any segmentation that had less than 20 instances was removed. the following list contains the data segmentations;

1. *Anesthetic method vs. Anesthetic type vs. Max BPM.*
2. *Max HR vs. Age.*
3. *Max HR vs. Anesthetic type*
4. *Max HR vs. Discomfort events*
5. *Max HR vs. Anesthesia Dosage*
6. *Anesthetic method vs. CHEOPS.*
7. *Anesthetic method vs. Discomfort events*
8. *Anesthetic method vs. Number of ampoules used vs. treated teeth.*
9. *Number of oxygen saturation dipping events vs. discomfort events vs. CHEOPS.*
10. *CHEOPS vs. Anesthesia dosage vs. Age.*

11. *Anesthesia dosage vs. Anesthetic method.*

12. *Treated tooth vs. Anesthetic method.*

13. *Treated tooth vs. Max HR.*

14. *Anesthetics method vs. duration of anesthesia injection.*

15. *Treatment duration vs. Anesthetic type.*

16. *Anesthetic method vs. Number of times additional anesthesia was given*

17. *Discomfort events vs. CHEOPS.*

The data segmentation was performed in MATLAB using the 'Database Explorer' add-on. This tool facilitates the connection to various databases, enabling efficient querying and direct importation of data into MATLAB. Notably, it allows these operations to be conducted without requiring extensive knowledge of SQL, thereby streamlining the data integration process. Once imported, the data was segmented as described in the previous section. This approach ensured that the segmentation process was both efficient and aligned with the study's analytical goals, reducing potential errors during data handling and preparation.

3.2.2 Normalized HR calculation

In pediatric patients, heart rate is known to vary significantly with age, making it a crucial factor to consider when analyzing physiological responses during dental procedures [44-46]. To account for these age-related variations, we introduced a parameter termed "Normalized heart rate". This parameter represents the difference between the maximum heart rate recorded during the dental procedure and the established normal resting heart rate for children of the same age as the patient which is then calculated based on the relative error formula as followed:

$$(1)\quad Normalized\ max\ heart\ rate\ [\%] = \frac{max\ heart\ rate\ [BPM] - established\ resting\ heart\ rate\ [BPM]}{established\ resting\ heart\ rate\ [BPM]}$$

3.2.3 Data Refining

The segments were refined separately to enhance data quality and reliability by the removal of outliers, empty data, and unreasonable values. Outliers, which are data points that deviate significantly from the rest of the dataset, can result from measurement errors, data entry mistakes, or genuine variability. Missing values are gaps

in the data that can compromise the accuracy and validity of the analysis if not properly addressed. Unreasonable values, defined as those falling outside expected ranges or logical boundaries, may indicate errors or anomalies and are excluded. These steps collectively improve the dataset's integrity, making it more suitable for accurate and meaningful analysis.

3.3 Statistical & Visual Analysis

Statistical analysis was performed for each segment to assess relationships and differences. T-test, Analysis of Variance (ANOVA) and two-way Analysis of Variance (ANOVA2) were used for comparing group means. In addition to these methods, correlation and regression analyses are essential for examining relationships between variables. Correlation measures the strength and direction of the linear relationship between two variables, typically quantified by the correlation coefficient, which ranges from -1 to 1. A positive value indicates a direct relationship, while a negative value suggests an inverse relationship. However, correlation does not imply causation. To explore predictive relationships, regression analysis is employed. Regression, particularly linear regression, models the relationship between a dependent variable and one or more independent variables, allowing researchers to predict the value of the dependent variable based on known values of the independent variables.

3.3.1 T-Tests

A t-test was employed to compare the means between two groups to determine if there was a statistically significant difference, under the assumption that the data is normally distributed. The T-tests were performed using the *ttest2()* function in Matlab. The significance level was set at 0.05, unless a Bonferroni correction was employed to adjust for multiple comparisons, where the adjusted significance level is given by formula (2):

$$(2) \qquad p_{adj} = \frac{0.05}{m}$$

with m being the number of comparisons [47] .

3.3.2 Analysis of Variance (ANOVA)

ANOVA was utilized to compare the means across three or more groups to determine if there were any statistically significant differences. This method is particularly effective in minimizing the increased risk of Type I errors that can arise from conducting multiple t-tests. By applying ANOVA, it is possible to assess whether the observed differences in group means are statistically significant, thereby providing insights into the relationships between the variables under investigation.

The Kruskal-Wallis test, a non-parametric equivalent to the one-way ANOVA, was employed to assess whether there were statistically significant differences in mean or median values of the dependent variable across multiple independent groups. This test is particularly useful when the assumptions of normality and homogeneity of variances are not met, as it does not require the data to follow a normal distribution. ANOVA and Kruskal-Wallis test were applied using the Matlab functions *anova() & kruskalwallis()*.

3.3.3 Two-Way Anova (ANOVA2)

Two-way ANOVA was employed to examine the effects of two independent variables on a dependent variable simultaneously and to assess any potential interaction between these variables. This statistical method allows for the evaluation of both the individual effects of each independent variable on the dependent variable, as well as their combined interaction. By using ANOVA2, researchers can gain a comprehensive understanding of how the independent variables influence the outcome, both independently and in conjunction with each other. Two-way ANOVA was applied using the Matlab function *anova2()*.

3.3.4 Visual Analysis

In addition to the statistical analysis, visual analysis was performed to allow for a more intuitive understanding of the data distribution, patterns, and relationships. Bar graphs were utilized to compare categorical data across different groups, providing a clear visualization of differences in means or counts, which complemented the t-test and ANOVA results. Heatmaps were employed to display the intensity of data points across a matrix, highlighting correlations or interactions between variables, especially useful in revealing patterns that might not be immediately obvious through numerical analysis alone. Scatterplots were used to visualize relationships between continuous variables, helping to identify trends, outliers, and the strength of associations, which was further quantified by correlation and regression analyses. These visual tools provided a comprehensive overview of the data, making it easier to communicate complex statistical findings and enhancing the overall interpretation of the study results.

4 Results

4.1 Segment 1: Anesthetic Method vs. Anesthetic Type vs. Max HR

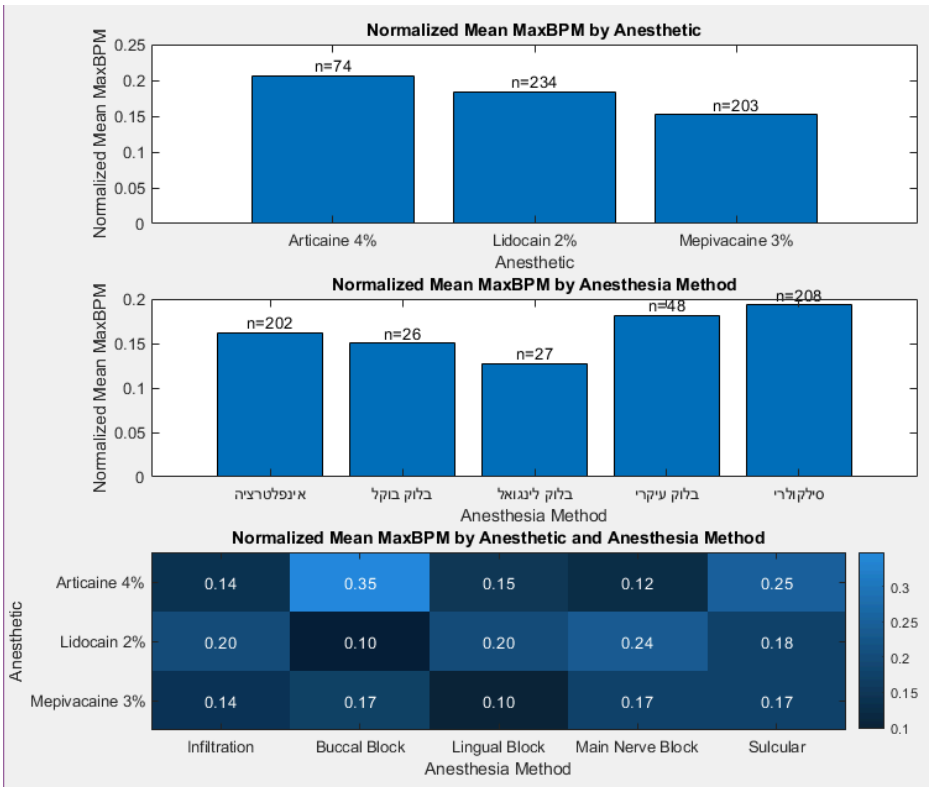

Figure 2: Normalized Mean Max BPM by Anesthetic Type and Anesthesia Method

A two-way ANOVA was conducted to assess the effects of anesthetic type and anesthesia method on maximum BPM, as well as their interaction. The results showed no significant effect of anesthetic type ($p = 0.5112$), anesthesia method ($p = 0.4502$), or their interaction ($p = 0.3239$) on maximum BPM.

The visual analysis is presented in Figure 2, which provides further insight into the effects of anesthetic type and anesthesia method on maximum BPM. Figure 2A shows that the normalized mean Max BPM was slightly higher for Articaine 4% compared to Lidocaine 2% and Mepivacaine 3%, although these differences were not statistically significant. Similarly, Figure 2B illustrates that the Buccal Block method exhibited a marginally higher normalized mean Max BPM compared to other methods, but no significant differences were detected. Additionally, the heatmap in Figure 2C reveals that the combination of Articaine 4% with Buccal Block resulted in the highest normalized mean Max BPM among the various combinations of anesthetic type and anesthesia method. However, consistent with the statistical analysis, no significant

variations were observed across the different groups. These visual representations support the ANOVA findings, indicating a lack of significant differences in Max BPM based on the variables studied.

4.2 Segment 2: Max HR vs. Age

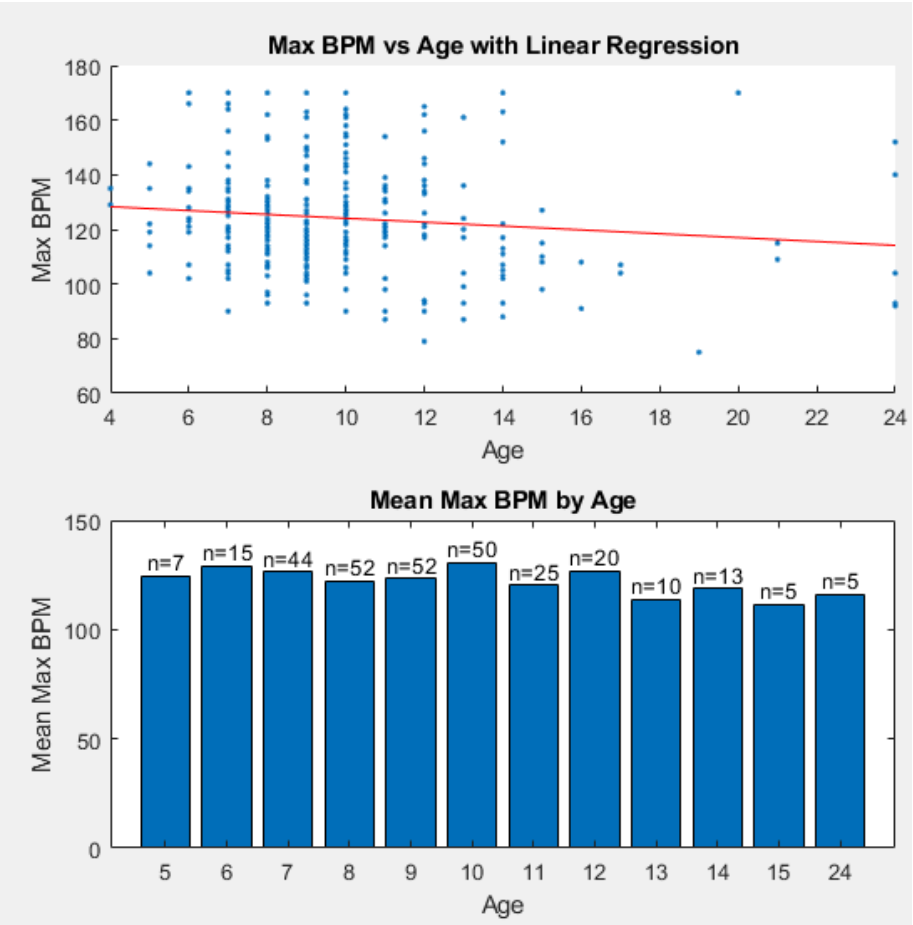

Figure 3: Scatter plot of Max BPM as a function of Age

A Pearson correlation analysis was conducted to evaluate the relationship between age and Max HR, yielding a Pearson correlation coefficient of -0.11 with a p-value of 0.0598. This suggests a weak negative correlation between age and Max BPM. The linear regression analysis produced a slope of -0.71 and an intercept of 131.14 [BPM], indicating a slight decrease in Max BPM with increasing age.

Figure 3 provides a visual representation of these findings. The scatter plot in Figure 3A shows the distribution of Max BPM across different ages, with the linear regression line reflecting the slight downward trend. Figure 3B presents the mean Max BPM for each valid age group, showing variations in mean Max BPM across different ages, with a general trend toward lower BPM values at older ages. The data for age

groups with at least 5 data points is included, showing the mean Max BPM for ages ranging from 5 to 24 years.

4.3 Segment 3: Max HR vs. Anesthetic Type

The maximum heart rate (HR) observed during dental procedures was analyzed for the three different anesthetics: Lidocaine 2%, Mepivacaine 3%, and Articaine 4%. The maximum HR for each treatment where the respective anesthetics were used was recorded and then averaged to generate the values depicted in (Figure 4).

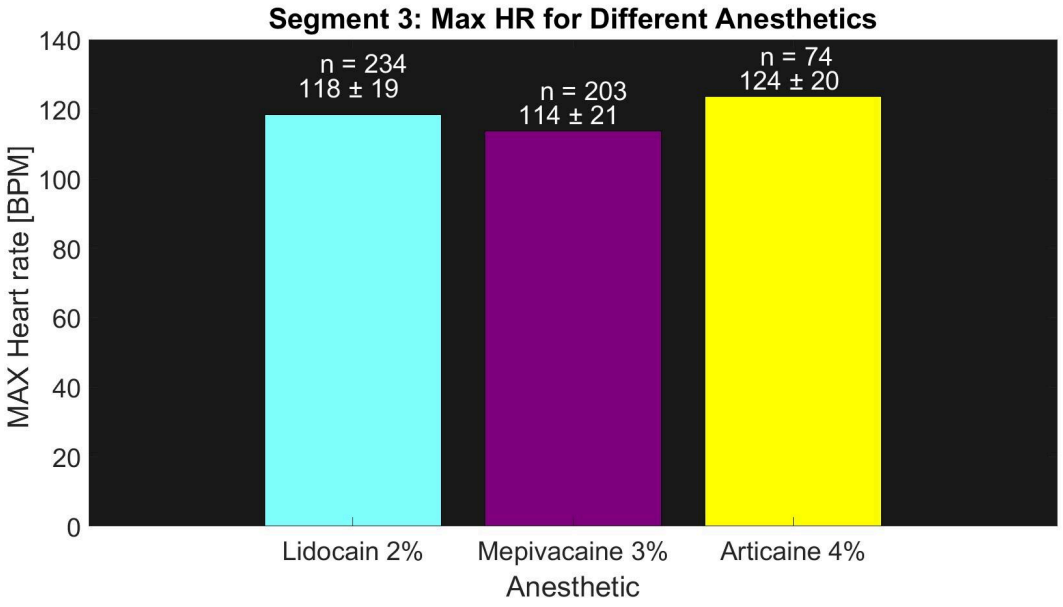

Figure 4: Bar plot of the average max HR [BPM] across the different anesthetics

The average maximum HR recorded was 124 ± 20 BPM for Articaine (n = 74), 118 ± 19 BPM for Lidocaine (n = 234), and 114 ± 21 BPM for Mepivacaine (n = 203).

A series of independent t-tests were conducted to determine if there were statistically significant differences in the maximum HR between the anesthetics. To account for multiple comparisons, a Bonferroni correction was applied (formula 2), setting the significance threshold at $P < 0.016$.

	Mepivacaine 3	Articaine 4%
Lidocaine 2%	p = 0.0132	p = 0.0504
Mepivacaine 3		p = 0.0005

Table 1: segment 3 max heart rate: pairwise t-test results with significant values highlighted ($p < 0.016$)

The results suggest that Mepivacaine 3% is associated with a significantly lower maximum HR compared to both Lidocaine 2% and Articaine 4%, while the difference between Lidocaine and Articaine does not reach statistical significance.

In addition, the normalized maximum HR observed during dental procedures was analyzed for the three different anesthetics: Lidocaine 2%, Mepivacaine 3%, and Articaine 4%. The maximum HR for each treatment where the respective anesthetics were used was recorded and then averaged to generate the values depicted in Figure 5.

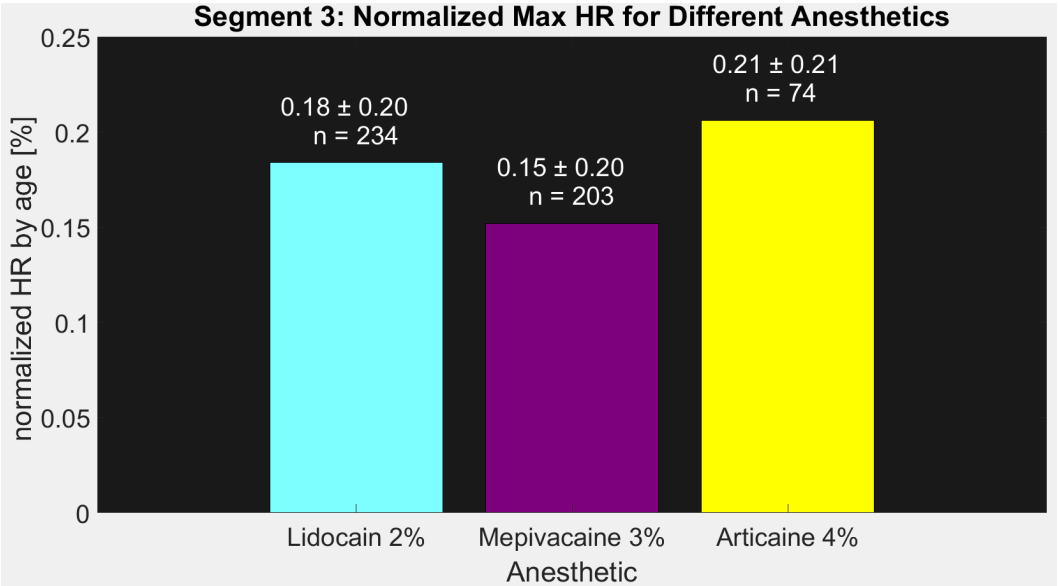

Figure 5: Bar plot of the average normalized max HR across the different anesthetics

The normalized HR values were 0.21 ± 0.21 BPM for Articaine (n = 74), 0.18 ± 0.20 BPM for Lidocaine (n = 234), and 0.15 ± 0.20 BPM for Mepivacaine (n = 203) (figure 5).

Pairwise t-tests were conducted to assess the significance of differences in age-normalized HR among the anesthetics. A Bonferroni correction was applied (formula 2) to adjust the significance threshold to $P < 0.016$.

	Mepivacaine 3	Articaine 4%
Lidocaine 2%	p = 0.0966	p = 0.4089
Mepivacaine 3		p = 0.0525

Table 2: segment 3 normed heart rate: pairwise t-test results with significant values highlighted ($p < 0.016$)

The results indicate that when normalizing HR by age, no statistically significant differences were observed between any pair of anesthetics after applying the Bonferroni correction. However, the Mepivacaine vs. Articaine comparison approached significance, suggesting a potential trend that could be explored in further studies with larger sample sizes.

4.4 Segment 4: Max HR vs. Signs of Discomfort

The maximum HR observed during dental procedures was analyzed across different types of discomfort experienced by patients. The discomfort types included Bitterness, Crying, Gagging, Hand Raising, Movement, and Pain. The overall average maximum HR across all treatments, regardless of discomfort, was 117 ± 20 BPM, which is represented by the black line in Figure 6 and table 3.

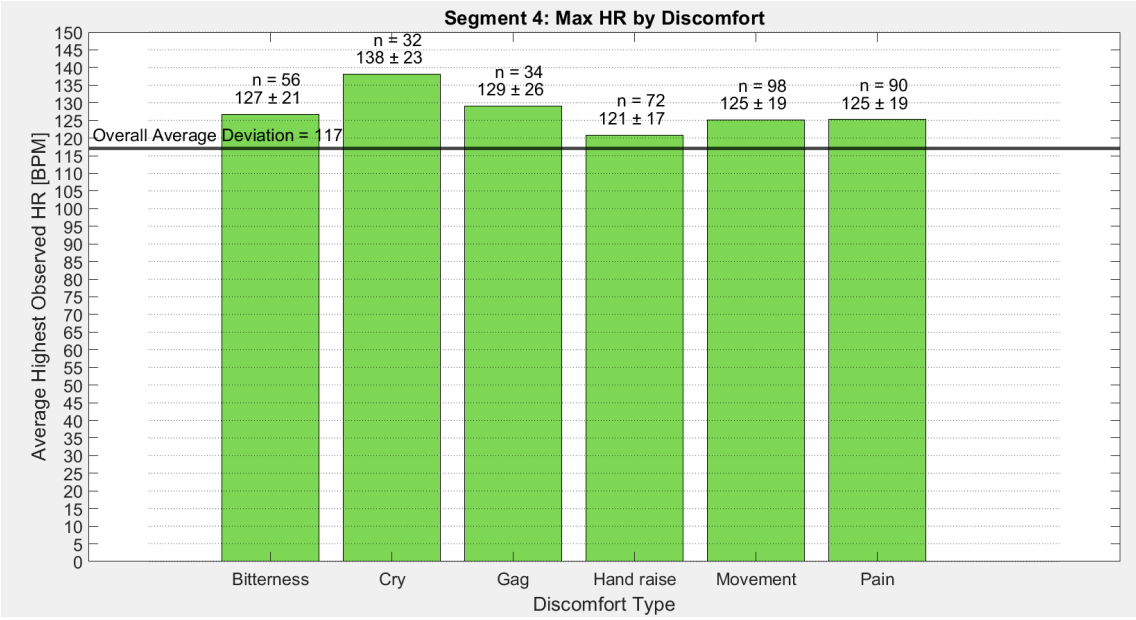

Figure 6: Bar Plot of Max HR as a function of Discomfort Type, With a Black Line Representing the Average Deviation of All Treatments.

	Mean [%]	STD [%]	N
Bitterness	127	21	56
Crying	138	23	36
Gagging	129	36	34
Hand raise	121	17	73
Movement	125	19	98
Pain	125	19	87
All Treatments	117	20	522

Table 3: segment 4 parameters

The data show that Crying is associated with the highest average maximum HR (138 [BPM]), followed by Gagging (129 [BPM]) and Bitterness (127 [BPM]). The overall average HR (117 [BPM]) serves as a baseline for comparison, and it can be seen that several discomfort types, particularly Crying and Gagging, result in higher HRs than the overall average.

A series of t-tests were performed to compare the maximum HR for each discomfort type against the overall average HR (117 BPM). The significance level was adjusted using the Bonferroni correction (formula 2), setting the significance threshold at $P < 0.008$.

	All Treatments
Bitterness	$p = 0.006$
Crying	$p < 0.001$
Gagging	$p < 0.001$
Hand raise	$p = 0.119$
Movement	$p = 0.002$
Pain	$p = 0.003$

Table 4: t-test results between each discomfort and the overall average with significant values highlighted ($p < 0.008$)

These results demonstrate that certain discomfort events, particularly Crying and Gagging, are associated with significantly higher maximum HRs compared to the overall average across all treatments. Bitterness, Movement, and Pain also showed significant increases in HR, while Hand Raising did not differ significantly from the overall average.

In addition, the normalized HR was analyzed across different types of discomfort experienced by patients. The discomfort types included Bitterness, Crying, Gagging, Hand Raising, Movement, and Pain. The overall average normalized

maximum HR across all treatments, regardless of discomfort, was 0.17 ± 0.20, which is represented by the black line in Figure 7 and table 5.

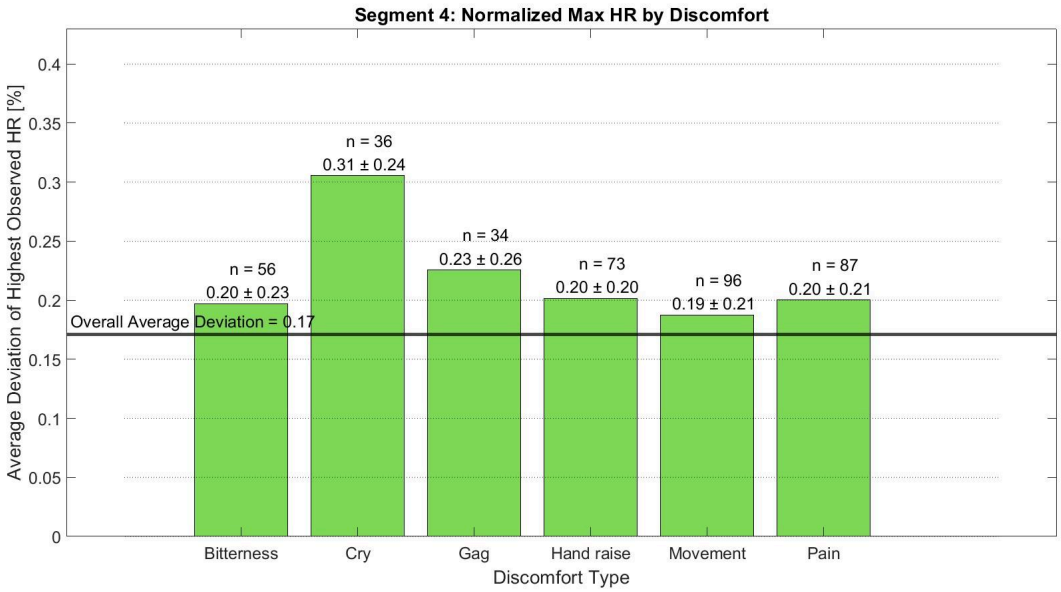

Figure 7: Bar Plot of Normalized Max HR as a function of Discomfort Type, With a Black Line Representing the Average Deviation of All Treatments.

	Mean [%]	STD [%]	N
Bitterness	20	23	56
Crying	31	24	36
Gagging	23	26	34
Hand raise	20	20	73
Movement	19	21	96
Pain	20	21	87
All Treatments	17	20	522

Table 5: segment 4 normalized HR parameters

The data indicates that Crying is associated with the highest average normalized maximum HR (0.31), followed by Gagging (0.23). The overall average normalized HR

(0.17) serves as a baseline for comparison, and it can be seen that Crying and Gagging result in higher normalized HRs than the overall average.

A series of t-tests were performed to compare the normalized maximum HR for each discomfort type against the overall average normalized HR (0.17). The significance level was adjusted using the Bonferroni correction (formula 2), setting the significance threshold at $P < 0.008$.

	All Treatments
Bitterness	$p = 0.002$
Crying	$p < 0.001$
Gagging	$p < 0.001$
Hand raise	$p = 0.370$
Movement	$p = 0.031$
Pain	$p < 0.001$

Table 6: t-test results between each discomfort and the overall average with significant values highlighted ($p < 0.008$)

These results demonstrate that certain discomfort events, particularly Crying, Gagging, and Pain, are associated with significantly higher normalized maximum HRs compared to the overall average across all treatments. Bitterness also showed a significant increase in normalized HR, while Hand Raising and Movement did not differ significantly from the overall average.

4.5 Segment 5: Max HR vs. Anesthesia Quantity

The relationship between the quantity of anesthesia administered and the deviation in normalized maximum heart rate (HR) was investigated using linear regression analysis. The scatter plot in Figure 8 illustrates this relationship, where each point represents an individual treatment, and the black line represents the linear fit.

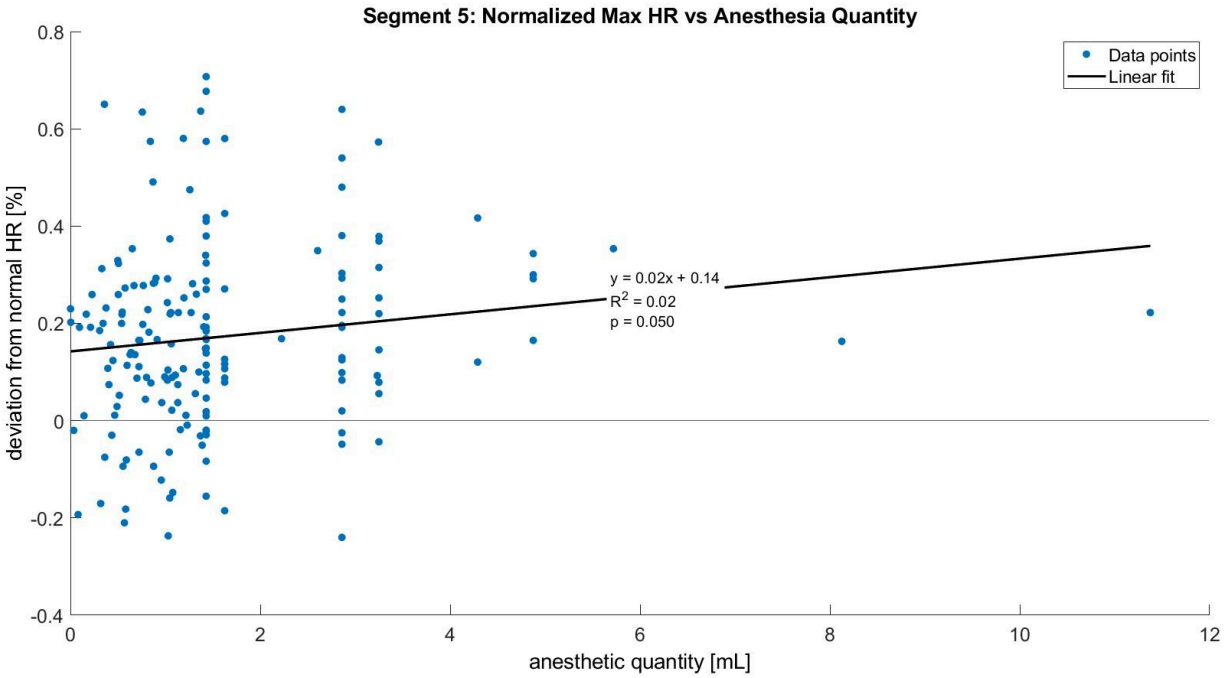

Figure 8: scatter plot showing the deviation of normalized maximum HR as a function of the quantity of anesthesia used during dental procedures

The data points are scattered widely, with a vertical spread showcasing the ampule volumes. Additionally, the Spearman correlation coefficient was calculated to be 0.1583, further supporting the weak association between these two variables.

These results suggest a statistically weak, positive linear relationship between the amount of anesthesia used and the deviation from the normalized maximum HR. While the p-value of 0.050 indicates that the relationship may be statistically significant, the low R^2 value and weak Spearman correlation suggest that this relationship is not practically significant. Thus, although there is a slight association between the two variables, it is weak and may not be clinically or practically meaningful.

4.6 Segment 6: Anesthetic Method vs. CHEOPS.

The CHEOPS scores were analyzed across three different anesthetic methods: Infiltration, Nerve Block, and Sulcular. The CHEOPS scores, which range from 5 to 13, were evaluated to determine the pain levels associated with each method. The data for CHEOPS scores are presented in Figures 9 and 10:

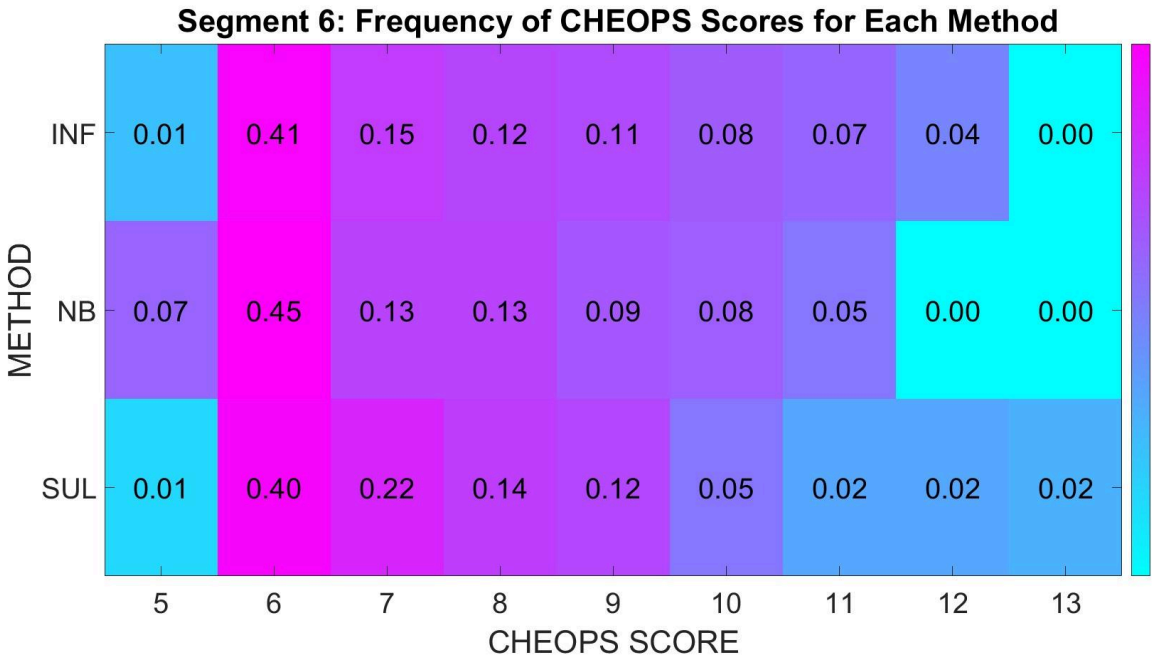

Figure 9: heatmap of the relative frequency of each CHEOPS score across the different anesthetic methods.

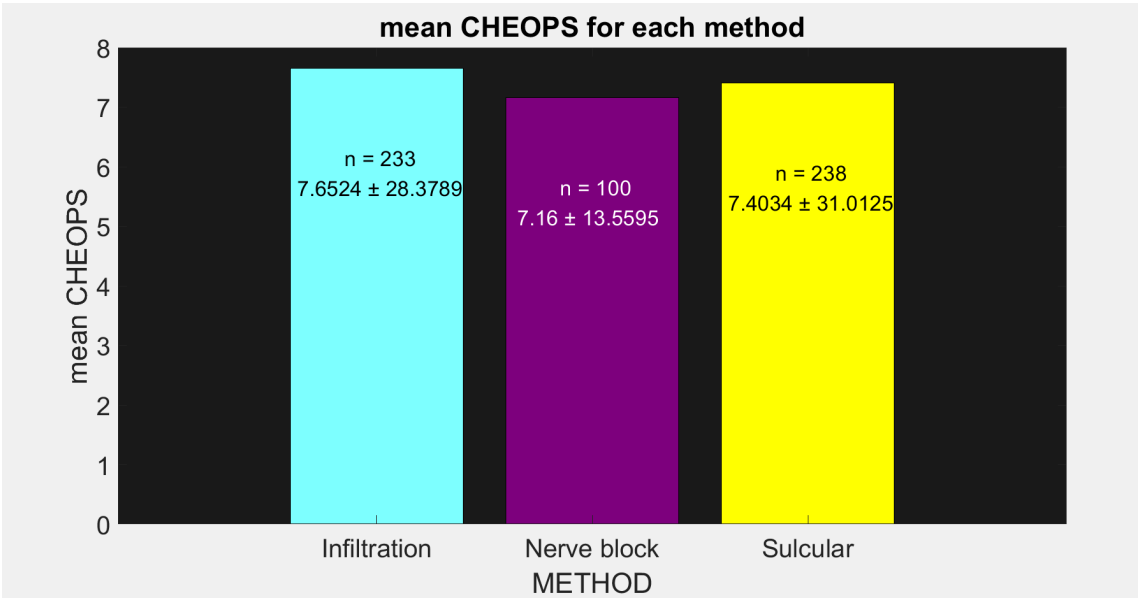

Figure 10: bar plot representing the mean CHEOPS scores for each method.

For all anesthetic methods, the highest frequency is observed for a CHEOPS score of 6, with a frequency of 0.41, 0.45, and 0.4 for infiltration, nerve block, and sulcular, respectively. The distributions show a decline in frequencies as the CHEOPS score increases, with instances of a recorded score of 13 only for the sulcular method. Notably, nerve block has a relatively higher frequency of scores at the lower end of the scale, particularly with a CHEOPS score of 5.

T-tests were conducted to compare the mean CHEOPS scores between the different anesthetic methods and their results can be seen in table 7:

	Nerve Block	Sulcular
Infiltration	P = 0.1852	P = 0.96887
Nerve Block		P = 0.20146

Table 7: Pairwise t-test results for each method

These results suggest that there are no significant differences in the CHEOPS pain scores among the different anesthetic methods. This indicates that the choice of anesthetic method does not significantly impact the level of pain as measured by the CHEOPS score.

4.7 Segment 7: Anesthesia Types vs. Discomfort events

The relative frequencies of various signs of discomfort were analyzed across three types of anesthesia: Lidocaine, Mepivacaine, and Articaine. The signs of discomfort considered included Bitterness, Crying, Gagging, Hand Raising, Movement, and Pain. The relative frequencies for each combination of anesthesia type and discomfort sign are presented in Figure 11:

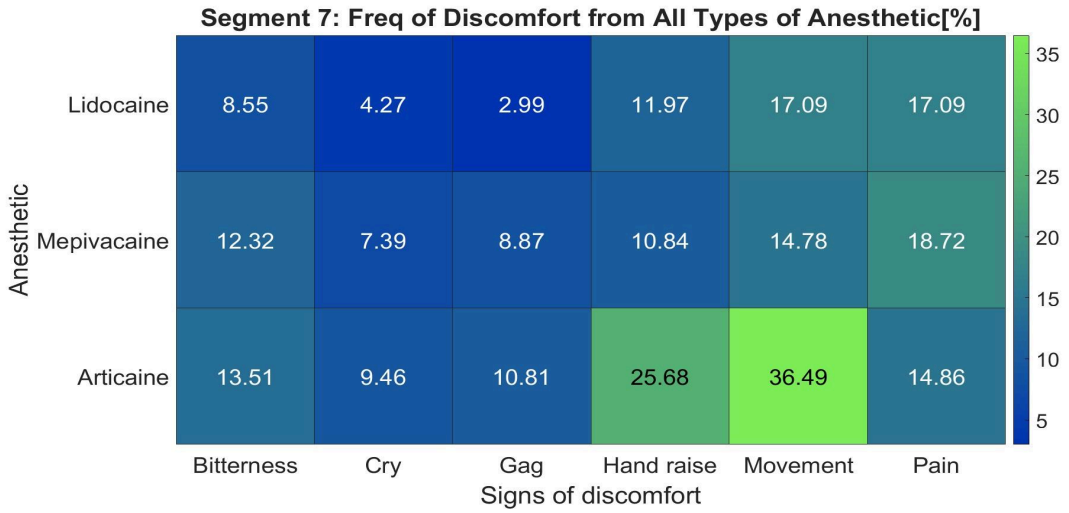

Figure 11: Heatmap of the relative frequencies of signs of discomfort, normalized by the total number of treatments for each anesthetic type

A series of pairwise t-tests were conducted to compare the proportions of various signs of discomfort across three anesthetic types: Lidocaine, Mepivacaine, and Articaine. The signs of discomfort analyzed include Bitterness, Crying, Gagging, Hand Raising, Movement, and Pain. A Bonferroni correction was applied (formula 2) to account for multiple comparisons, setting the significance threshold at $P<0.0033$.

Lidocaine showed significantly lower bitterness occurrences compared to Articaine and mepivacaine, while all comparisons between the other signs were statistically significant, suggesting that the type of anesthetic used has a substantial impact on the occurrence of these discomfort signs. These results highlight the importance of considering the choice of anesthetic in relation to the likelihood of discomfort during dental procedures.

4.8 Segment 8: Anesthetic method vs. Number of ampoules used vs. Treated teeth.

The average number of anesthesia ampules used during dental procedures was analyzed across various teeth and anesthetic methods, including Infiltration, Buccal Block, Lingual Block, Main Block, and Sulcular techniques. Figure 12 presents a heatmap visualizing the ampule usage for each tooth-method combination, while Figure 13 represents the number of instances for each tooth-method combination.

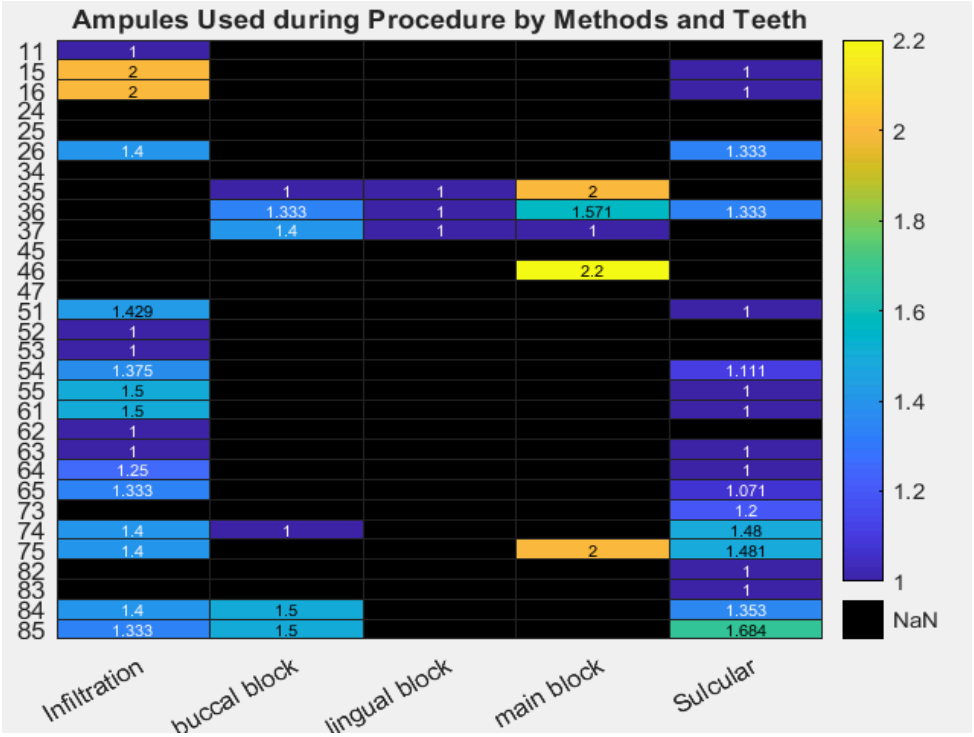

Figure 12: heatmap where each row corresponds to a specific tooth, identified by its number on the left, and each column represents an anesthetic method. The numbers within the cells indicate the average number of ampules used for the respective tooth and method.

	Infiltration	Buccal Block	Lingual Block	Main Block	Sulcular
Tooth 11	1	0	0	0	0
Tooth 15	1	0	0	0	1
Tooth 16	4	0	0	0	1
Tooth 24	0	0	0	0	0
Tooth 25	0	0	0	0	0
Tooth 26	5	0	0	0	3
Tooth 34	0	0	0	0	0
Tooth 35	0	1	1	2	0
Tooth 36	0	9	4	7	3
Tooth 37	0	5	1	3	0
Tooth 45	0	0	0	0	0
Tooth 46	0	0	0	5	0
Tooth 47	0	0	0	0	0
Tooth 51	7	0	0	0	2
Tooth 52	1	0	0	0	0
Tooth 53	2	0	0	0	0
Tooth 54	8	0	0	0	9
Tooth 55	14	0	0	0	6
Tooth 61	6	0	0	0	1
Tooth 62	1	0	0	0	0
Tooth 63	1	0	0	0	1
Tooth 64	12	0	0	0	9
Tooth 65	15	0	0	0	14
Tooth 73	0	0	0	0	5
Tooth 74	5	1	0	0	25
Tooth 75	5	0	0	1	27
Tooth 82	0	0	0	0	1
Tooth 83	0	0	0	0	1
Tooth 84	5	2	0	0	17
Tooth 85	6	2	0	0	19

Figure 13: the number of instances for each tooth-method combination

Tooth 46 shows the highest ampule usage, with an average of 2.2 ampules under the Main Block method. In contrast, the Infiltration method for tooth 15 and the Main Block method for tooth 15 each show an average of 2 ampules used. The heatmap also includes several black cells marked as 'NaN,' indicating missing data or tooth-method combinations for which no procedures were recorded.

Although the heatmap in Figure 12 offers valuable insights into ampule usage patterns across various anesthetic methods and teeth, the limitations in data availability, small sample sizes, and data heterogeneity hinder the ability to perform a robust statistical analysis. To draw more reliable conclusions, a larger dataset with more evenly

distributed data would be required. This would enable more meaningful statistical comparisons across different anesthetic methods and tooth locations.

4.9 Segment 9: Number of Oxygen Saturation Drops vs. Cheops Pain Scale

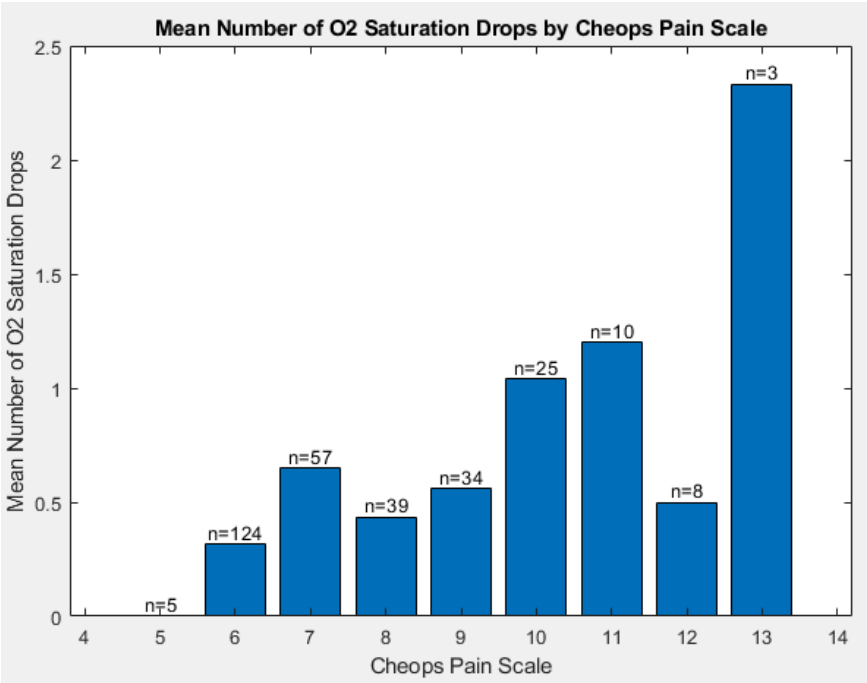

Figure 14: Average number of O2 saturation drops during procedure as a function of Cheops Pain scale

A Kruskal-Wallis ANOVA was conducted to evaluate the differences in the number of O2 saturation drops across various levels of the Cheops Pain Scale. The results show a statistically significant difference among the groups (Chi-square = 23.69, p = 0.0026).

Figure 9 illustrates the mean number of O2 saturation drops corresponding to different Cheops Pain Scale scores. The bar graph depicts how the average number of O2 saturation drops varies with increasing Cheops Pain Scale scores. The sample size for each pain scale score is provided above the respective bars, indicating the distribution of data points across the different pain levels.

4.10 Segment 10: Cheops, Anesthesia Quantity, & Age

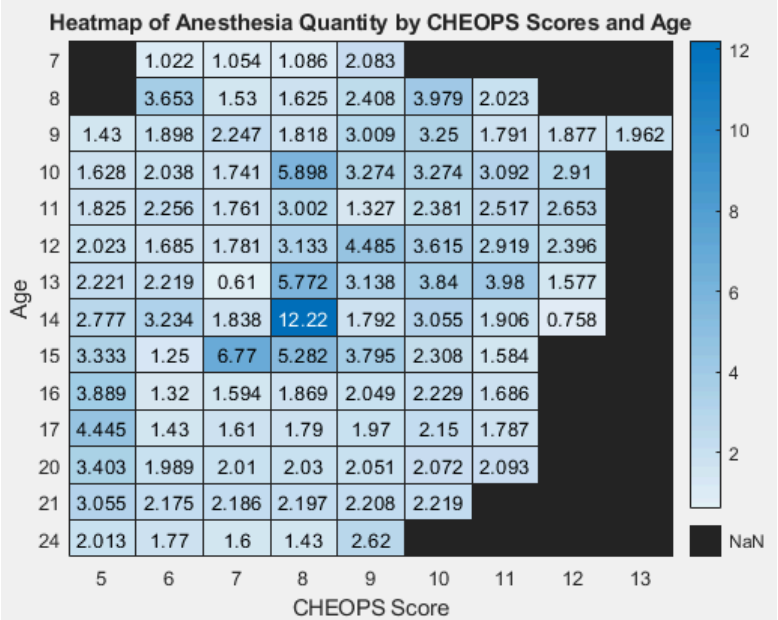

Figure 15: Heatmap of Anesthesia Quantity [ml] by CHEOP scores and age

Figure 15 presents a visualization exploring the relationship between Cheops pain scores, anesthesia quantity, and patient age. The heatmap illustrates the quantity of anesthesia administered (in milliliters) across varying Cheops scores and ages. The heatmap shows how the amount of anesthesia correlates with different pain levels and age groups, with each cell representing the average anesthesia quantity corresponding to a specific combination of Cheops score and age. The color gradient highlights the variations in anesthesia quantity, with lighter colors indicating higher volumes administered.

4.11 Segment 11: Anesthesia dosage vs. Anesthetic method.

The mean quantity of anesthesia used was analyzed across different anesthetic methods, including infiltration, buccal block, lingual block, main block, and sulcular techniques. The overall average quantities of anesthesia used for each method are presented in Figure 16 and Table 8.

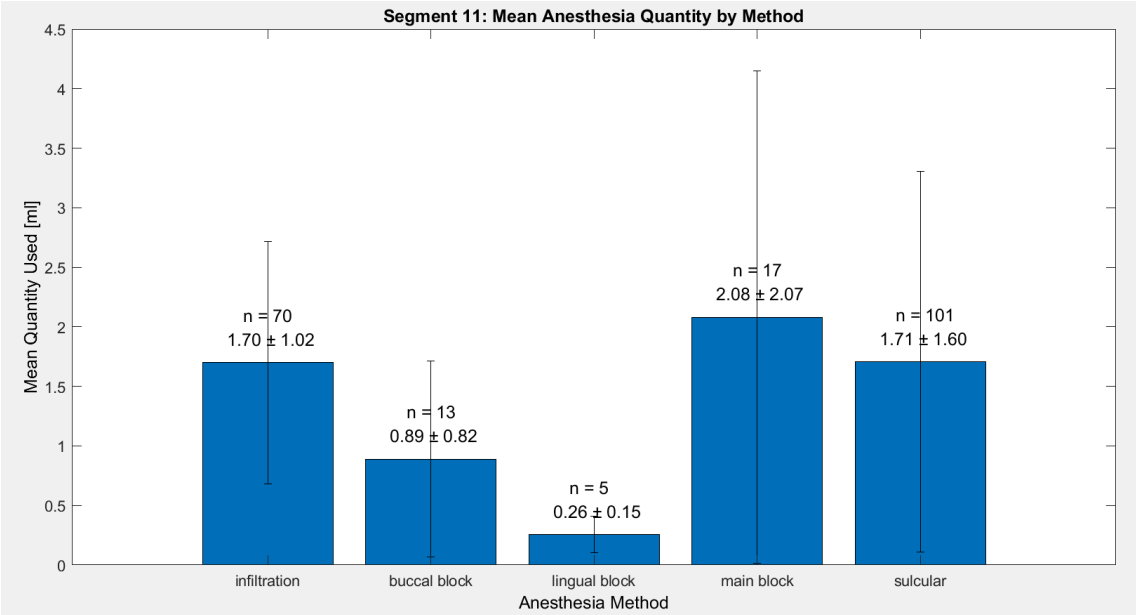

Figure 16: bar plot representing the mean quantity of anesthesia administered for each anesthetic method.

	Mean [ml]	STD [ml]	N
Infiltration	1.7	1.02	70
Buccal block	0.89	0.82	13
Lingual block	0.26	0.15	5
Main block	2.08	2.07	17
Sulcular	1.71	1.6	101

Table 8: segment 11 parameters

The infiltration method had a mean anesthesia quantity of 1.70 ± 1.02 [mL] (n = 70), while the buccal block and sulcular methods used mean quantities of 0.89 ± 0.82 [mL] (n = 13) and 1.71 ± 1.60 mL (n = 101), respectively. The main block method had the highest mean quantity at 2.08 ± 2.07 mL (n = 17), whereas the lingual block method used the least amount of anesthesia, with a mean quantity of 0.26 ± 0.15 mL (n = 5). While the means differ across methods, the variability within each group is relatively high, as reflected in the large standard deviations.

A series of pairwise t-tests were conducted to compare the mean anesthesia quantities between different anesthetic methods. The significance level was adjusted using the Bonferroni correction (formula 2), setting the significance threshold at $P < 0.005$. The results are summarized in Table 9.

	Buccal Block	**Lingual Block**	**Main Block**	**Sulcular**
Infiltration	p = 0.0082	p = 0.0023	p = 0.2736	p = 0.9695
Buccal Block		p = 0.1126	p = 0.0605	p = 0.0727
Lingual Block			p = 0.0668	p = 0.0456
Main Block				p = 0.3949

Table 9: segment 11 pairwise t-test results with significant value (p < 0.005) highlighted

These results suggest that after applying the Bonferroni correction, the infiltration method requires significantly more anesthesia than the lingual block method, with infiltration and buccal block being close to significant. The other comparisons did not show significant differences in the quantities of anesthesia used at the corrected significance level.

4.12 Segment 12: Treated Tooth vs. Anesthesia Method

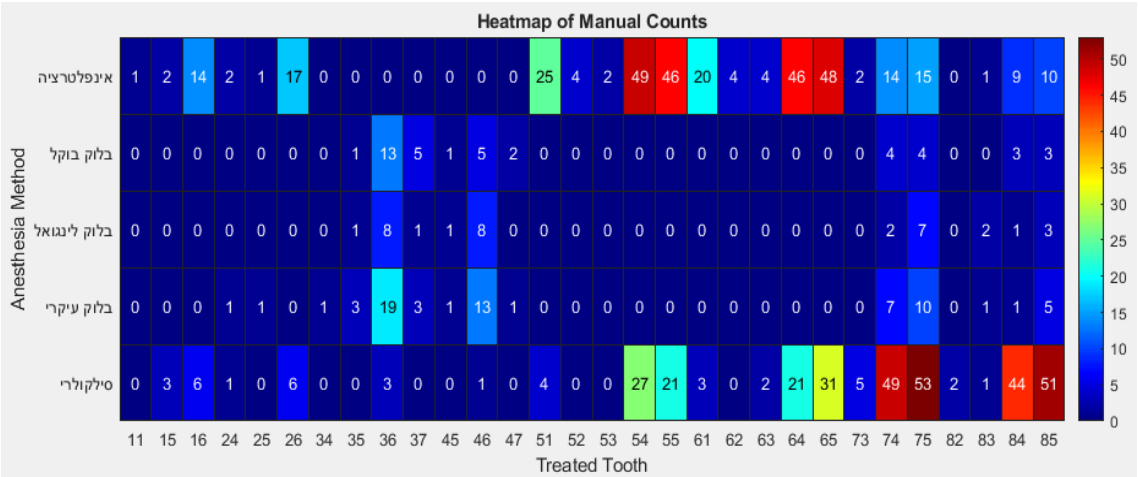

Figure 17: Heatmap of frequency of treated tooth (numbered) by anesthesia method

The heatmap presented in Figure 17 visualizes the frequency of treated teeth in relation to different anesthesia methods. Each cell in the heatmap represents the number of instances a particular tooth was treated using a specific anesthesia method. The treated teeth are numbered along the x-axis, while the anesthesia methods are listed on the y-axis. The color gradient indicates the frequency, with warmer colors (e.g., red) representing higher frequencies and cooler colors (e.g., blue) representing lower frequencies.

4.13 Segment 13: Treated Tooth vs. Max HR

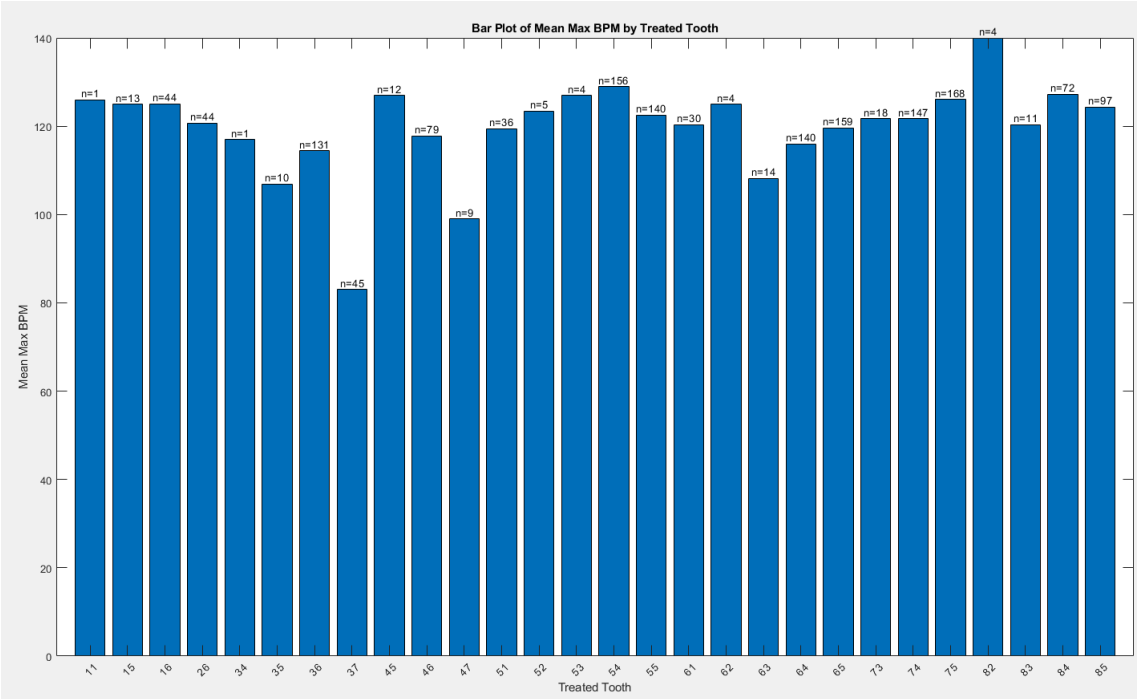

Figure 18: Bar graph of each tooth number and its associated mean HR

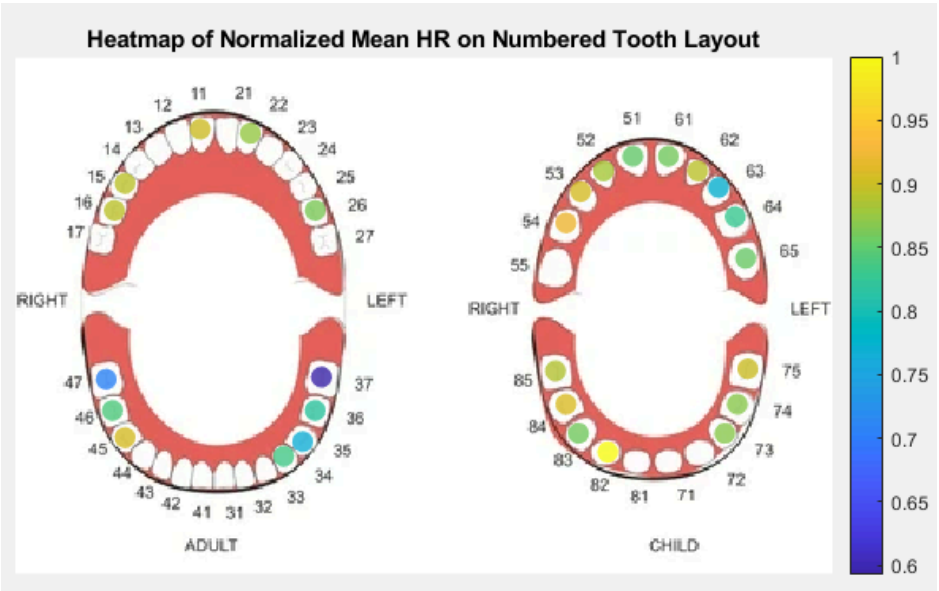

Figure 19: Heatmap overlay of normalized average HR per tooth treated

Figures 18 and 19 include a bar graph and a heatmap that visualize the relationship between treated tooth numbers and their associated mean max HR.

The bar graph displays the mean max HR for each treated tooth number, with each bar representing the mean max HR recorded for that specific tooth. The height of the bars indicates the mean max HR values for different teeth, facilitating comparison.

The heatmap provides an overlay of the normalized mean max HR on a dental layout, differentiating between adult and child teeth. The color gradient indicates the normalized mean max HR values, with variations in color representing different HR levels.

Additionally, a Spearman's rank correlation analysis was conducted to assess the relationship between the treated tooth and Max HR. The analysis yielded a Spearman's rank correlation coefficient of -0.19699 with a p-value of 1.8569e-06, indicating a statistically significant correlation between the treated tooth and Max HR.

4.14 Segment 14: Anesthetics method vs. Anesthesia duration

The average duration of anesthesia was analyzed across different anesthetic methods, including infiltration, buccal block, lingual block, main block, and sulcular techniques. The average durations for each method are presented in Figure 20 and Table 10.

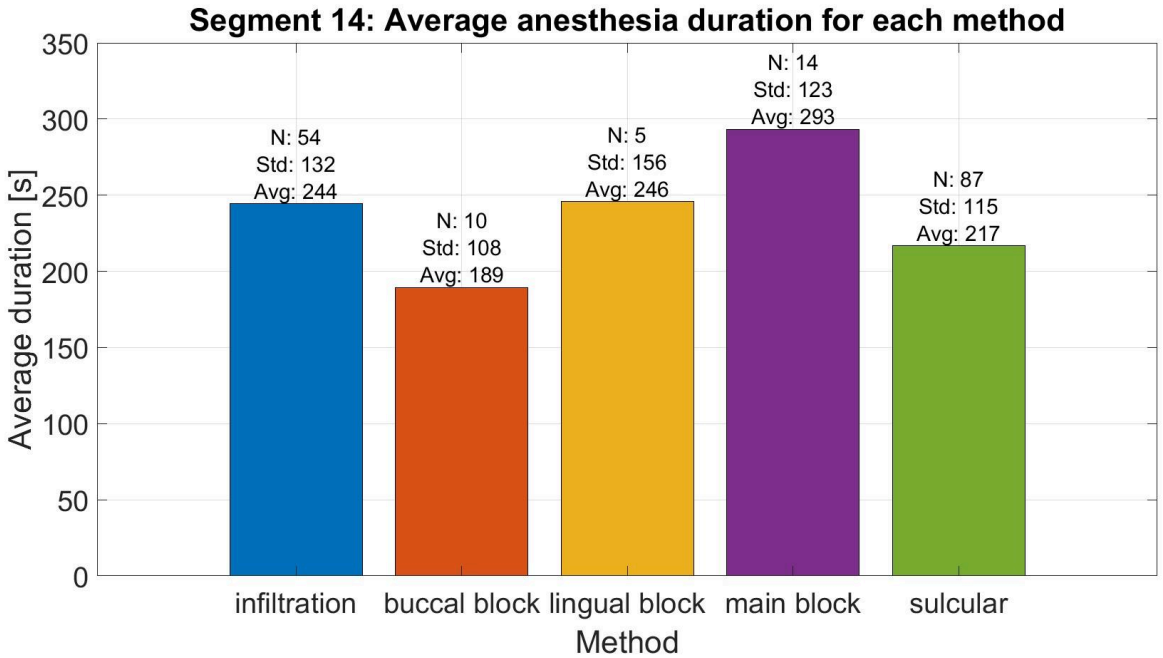

Figure 20: bar plot representing the average duration of anesthesia for each anesthetic method.

	Mean [s]	STD [s]	N
Infiltration	244	132	54
Buccal Block	189	108	10
Lingual Block	246	156	5
Main Block	293	123	14
Sulcular	217	115	87

Table 10: segment 14 parameters

The main block method had the longest average duration of 293 ± 123 seconds (n = 14), followed by the lingual block with 246 ± 156 seconds (n = 5) and the infiltration method with 244 ± 132 seconds (n = 54). The buccal block method had the shortest average duration of 189 ± 108 seconds (n = 10), while the sulcular method had an average duration of 217 ± 115 seconds (n = 87).

A series of pairwise t-tests were conducted to compare the average anesthesia durations between different anesthetic methods. The significance level was adjusted using the Bonferroni correction (formula 2), setting the significance threshold at $P < 0.005$. The results are summarized in table 11.

	Buccal Block	Lingual Block	Main Block	Sulcular
Infiltration	p = 0.1753	p = 0.9790	p = 0.2047	p = 0.2181
Buccal Block		p = 0.4908	p = 0.0397	p = 0.4562
Lingual Block			p = 0.5650	p = 0.7021
Main Block				p = 0.0451

Table 11: segment 14 pairwise t-test results with significant value ($p < 0.005$) highlighted

These results suggest that after applying the Bonferroni correction, no significant differences in anesthesia duration were found between the different anesthetic methods.

4.15 Segment 15: Treatment Duration vs. Anesthetic Type

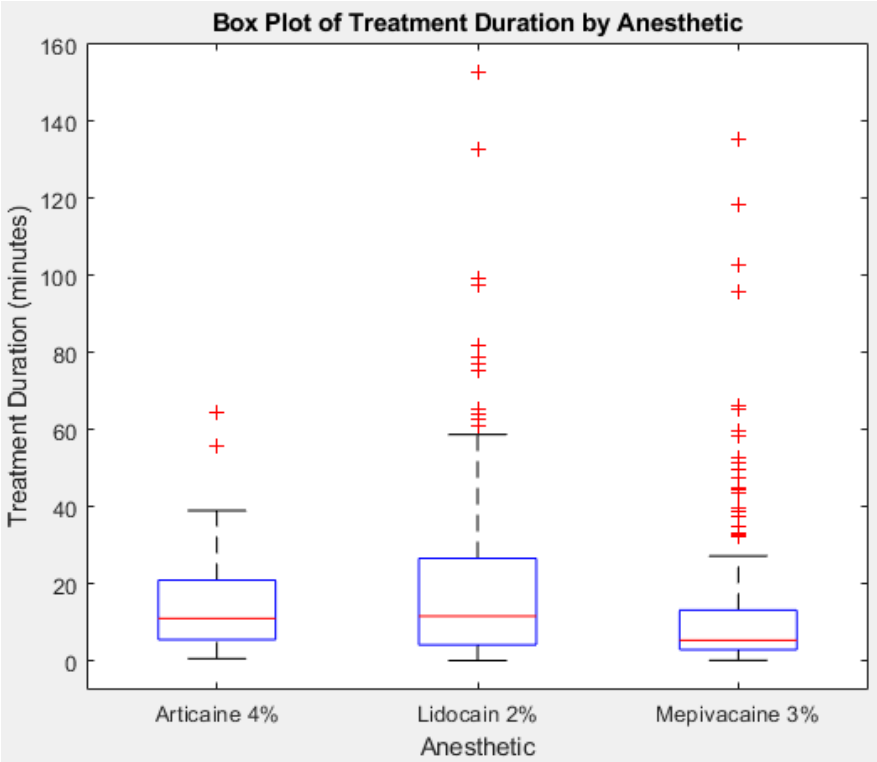

Figure 21: Box plot of treatment duration per anesthetic type

A Spearman's rank correlation analysis was conducted to evaluate the relationship between treatment duration and anesthetic type. The analysis produced a Spearman's rank correlation coefficient of -0.19699 with a p-value of 1.8569e-06, suggesting a correlation between these variables.

Figure 21 presents a box plot that visualizes the distribution of treatment durations across different anesthetic types (Articaine 4%, Lidocaine 2%, and Mepivacaine 3%). The box plot displays the median, interquartile range, and any outliers for each anesthetic type, allowing for a comparison of treatment durations

associated with each anesthetic. The spread of the data and the presence of outliers are indicated by the length of the whiskers and the points outside the whiskers.

4.16 Segment 16: Anesthesia Restarts vs. Anesthesia Method

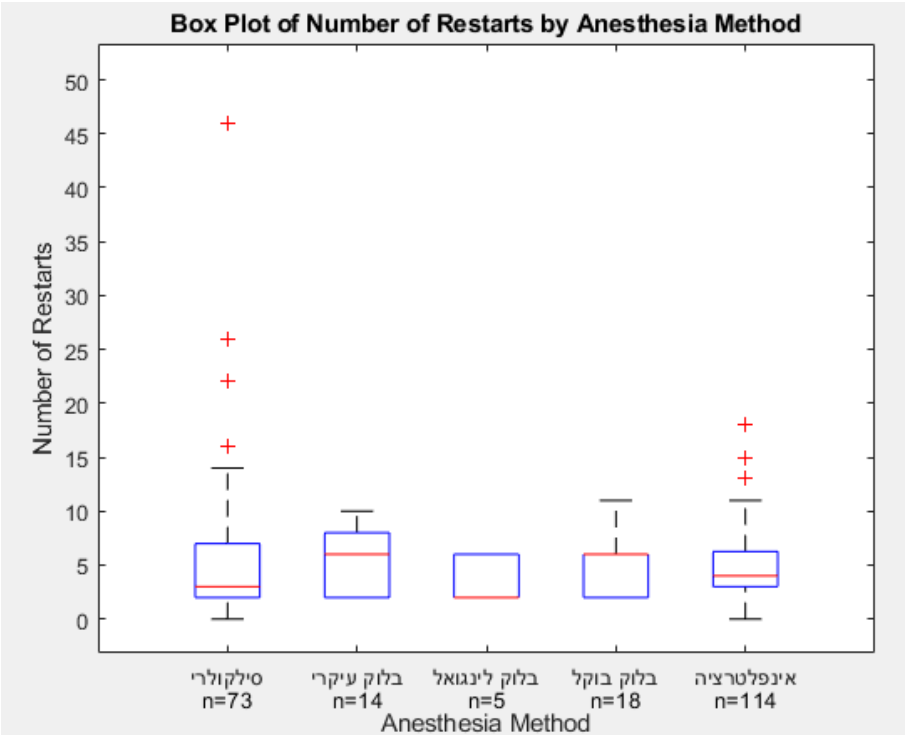

Figure 22: Box plot of average number of Anesthesia restarts per each anesthesia method

A Kruskal-Wallis ANOVA was performed to assess differences in the number of anesthesia restarts across various anesthesia methods. The results (Chi-square = 4.21, p = 0.3777) indicate that there is no statistically significant difference among the groups.

Figure 22 provides a box plot illustrating the distribution of the number of anesthesia restarts for each anesthesia method. The plot shows the median, interquartile range, and outliers for each method, allowing for a comparison of the frequency of restarts across the different methods. The sample size for each anesthesia method is indicated below the corresponding box.

4.17 Segment 17: Discomfort events vs. Cheops Pain Scale

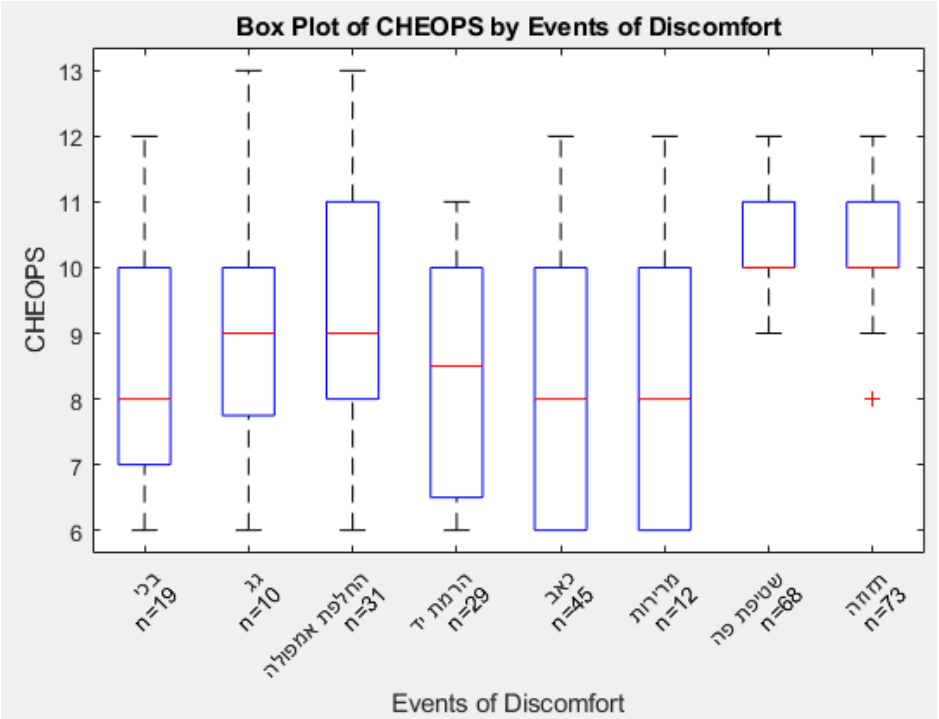

Figure 23: Box plot of Cheops pain scale scores for each event of discomfort

A Kruskal-Wallis ANOVA was performed to examine the differences in Cheops pain scale scores across various events of discomfort. The results indicate a statistically significant difference among the groups (Chi-square = 33.56, p = 2.0855e-05).

Figure 23 provides a box plot that visualizes the distribution of Cheops pain scale scores for each event of discomfort. The box plot displays the median, interquartile range, and outliers for each event category, with the number of observations (n) indicated below each event label. This figure allows for a comparison of pain scale scores across different discomfort events, illustrating the variation in pain levels associated with each event type.

5 Discussion & Conclusions

This study encountered several challenges due to the sparse nature of the dataset, despite the initial appearance of a large database. The segmentation of data across numerous variables, such as anesthetic type, patient age, and treated tooth, resulted in some segments with fewer than 20 data points, limiting the robustness of the statistical analysis. Additionally, certain variables were inherently limited, further constraining the conclusions that could be drawn. For example, the age range in the dataset was somewhat restricted, predominantly covering children aged 5 to 15, with some age groups being underrepresented. This limitation may have affected our ability to fully explore age-related trends and their impact on pain management and anesthetic response.

Despite these limitations, the results of this analysis provide valuable insights for developing pediatric dental protocols. For instance, Segment 1 examined maximum heart rate (HR) in relation to different anesthetic types and methods, revealing subtle differences in normalized max HR, though these were not statistically significant. Segment 13, which explored the relationship between treated tooth number and associated max HR, underscores the importance of considering physiological responses to different dental procedures, potentially aiding in the selection of anesthetic methods that minimize stress on the cardiovascular system. Segment 7's exploration of discomfort events in relation to anesthesia types further emphasized the significance of anesthetic choice, showing varying frequencies of discomfort events like crying and gagging, depending on the anesthetic used. This insight is crucial for tailoring anesthesia protocols that minimize patient discomfort and stress during dental procedures.

Moreover, the study highlighted significant findings regarding the influence of procedural factors on anesthesia management. While Segment 16 suggested that certain anesthesia methods might be associated with higher rates of restarts during dental procedures, the differences between these methods were visible but not major or statistically significant. This implies that although variations in anesthesia methods exist, they may not have a substantial impact on the stability and effectiveness of anesthesia in a clinical setting. In contrast, Segment 11 explored the connection between anesthesia dosage and anesthetic method, showing that the amount of anesthesia required varied significantly across different methods. This variability could reflect differences in the effectiveness or stability of each method, particularly in maintaining adequate anesthesia over time.

Segment 2 examined the relationship between patient age and maximum HR, revealing a slight but significant negative correlation. As patient age increased, Max HR tended to decrease, highlighting the importance of considering age-related physiological changes when administering anesthesia. Segment 10 extended this analysis by exploring the correlation between CHEOPS pain scores and the quantity of anesthesia administered, showing a significant association between higher pain scores and increased anesthesia doses. These findings suggest that both age and pain levels should be carefully monitored and factored into the anesthetic protocol to optimize patient comfort and safety.

The analysis of Segment 14 focused on the duration of anesthesia across different anesthetic methods, revealing notable variations in average anesthesia duration. These findings suggest that while there may be practical differences in how long different anesthetic methods maintain their effects, these differences are not substantial enough to warrant significant clinical concern based on the current dataset. However, the variability in duration highlights the importance of monitoring anesthesia duration during pediatric dental procedures to ensure that the selected method provides sufficient anesthesia throughout the treatment, without necessitating additional doses or risking prolonged exposure.

The analysis in Segment 4 of HR responses to different signs of discomfort, such as crying, gagging, and movement, revealed that these discomfort signals were associated with significant fluctuations in heart rate. This suggests that monitoring for specific signs of discomfort can be critical in anticipating and managing stress-related cardiovascular responses in pediatric patients. By identifying which discomfort signs are most likely to provoke a strong physiological reaction, dental practitioners can implement preemptive strategies—such as adjusting the anesthesia dosage or providing additional sedation—to mitigate stress and prevent adverse cardiovascular outcomes.

Following this, Segment 5 explored the correlation between HR and anesthesia dosage, suggesting a possible connection between the quantity of anesthesia administered and the physiological stress experienced by pediatric patients. However, the results from this segment were not critical, with a low R-squared value and a weak Spearman correlation. This indicates that while there may be some relationship between anesthesia dosage and HR, it is not statistically significant or strongly predictive. This conclusion aligns with the results from Segment 11, which demonstrated variability in anesthesia dosage requirements across different methods. Although the findings from Segment 5 are less robust, the combined insights from these segments still emphasize the importance of selecting anesthesia methods that require precise and consistent dosing while also effectively managing physiological responses, such as heart rate. By carefully choosing and administering the appropriate anesthetic dosage, dental practitioners can work towards safer outcomes for pediatric patients during dental procedures.

Finally, findings from Segment 13 highlighted a significant variation in the maximum HR associated with different treated teeth, as visualized through both the bar graph (figure and heatmap. The findings revealed patterns in HR response, suggesting that certain teeth may be associated with higher physiological stress during dental procedures. This underscores the importance of tailoring anesthesia approaches, not just based on the type of procedure, but also considering the specific tooth being treated, to enhance patient safety and comfort during dental treatments.

By integrating findings from different segments, this study provides a comprehensive understanding of the complex interactions between biological markers and medical interventions in pediatric dentistry, which is crucial for developing more effective and safer anesthesia protocols tailored to the unique needs of young patients. However, the limitations encountered, particularly the sparse datasets, underscore the need for further data collection and analysis. Expanding and diversifying the database will be essential for developing a more robust protocol for pediatric dental anesthesia and drawing stronger conclusions. Nonetheless, the data analysis presented here offers valuable tools for dentists and researchers, providing deeper insights into the connections between biological markers and anesthesia in pediatric dentistry. These insights may guide future research and clinical practices aimed at enhancing patient safety and comfort during dental procedures.

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